iv. Role in gastro-intestinal function:

Carbohydrates play an important role in the gastro-intestinal functions of mammals. The digestive system changes carbohydrates into glucose, also known as blood sugar. Some glucose is used for energy and the rest is stored in the liver and muscles for later use. When blood sugar rises, pancreas pumps out more and more insulin, a hormone that tells cells to absorb glucose for energy or storage. As cells absorb wwglucose, blood sugar levels begin to fall, which signals the pancreas to start making glucagon, a hormone that tells the liver to release stored glucose.

Anticoagulants, commonly referred to as blood thinners, are chemical substances that prevent or reduce coagulation of blood, prolonging the clotting time.

v. Carbohydrate functions as Antigen:

Many antigens are glycoproteins (which contain oligosaccharide) in nature and give immunological properties to the blood.

vi. Carbohydrate functions as Hormone:

Many Hormones like FSH (Follicular Stimulating Hormone which takes part in ovulation in females) and LH (Leutinizing Hormone) are glycoprotein and help in reproductive processes.

vii. Carbohydrates provide raw material for industry:

Carbohydrates are an important component of many industries like textile, paper, lacquers and breweries.

viii. Other Functions

Agar is polysaccharide used in culture media, laxative and food.

Cellulose acts as roughage of food. It stimulates peristaltic movement and in the secretion of digestive enzymes.

Hyaluronic acid found between joints acts as synovial fluid and provides frictionless movement.

Unit 5.indd 123 21-05-2018 15:10:27